

POLITECNICO DI MILANO







The firm and the context: the demand

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What determines a firm's profits?

The profits earned by a firm depend on:

1. demand for the product or service (good) marketed by the firm
2. competition
3. interactions between the firm and other stakeholders

Market

The market

Definition of market

In economics, a market is the place (not necessarily physical) where exchanges of goods and services take place at given prices.

The market is also the place where demand and supply of specific goods meet, that is, where buyers and sellers interact economically.

Elements of the market

In every market it is possible to identify:

– buyers' demand

– sellers' supply

– price

The demand

The classical theory of demand focuses on purchasing decisions related to final goods (not intermediate goods).

Demand is the quantity of a good that buyers are willing to purchase at a given price.

Several factors affect demand, but the most important is price.
Other factors include consumers' tastes (fashion, social trends), the price of other goods, and income.

Who are consumers?

Consumers:

– economic agents who purchase and consume goods in order to increase their well-being (utility)

– economic agents willing to pay to purchase goods

Consumers' preferences are summarised by the utility function, a measure of the satisfaction derived from purchasing and consuming goods.

Consumer behaviour

Consumers maximise their utility function subject to their budget constraints, meaning that expenditure on goods cannot exceed income.

Fundamental assumption: consumers are perfectly rational economic agents, they know all goods available on the market and their own income.

The individuals' demand

- Individual demand for a good depends on:
  – the price of the good
  – the price of substitute or complementary goods
  – individual preferences
  – individual income

Consumers have a finite income to allocate in order to satisfy a number N of needs.

Reservation price

Reservation price (RP): the maximum price a consumer is willing to pay to purchase one unit of a good.

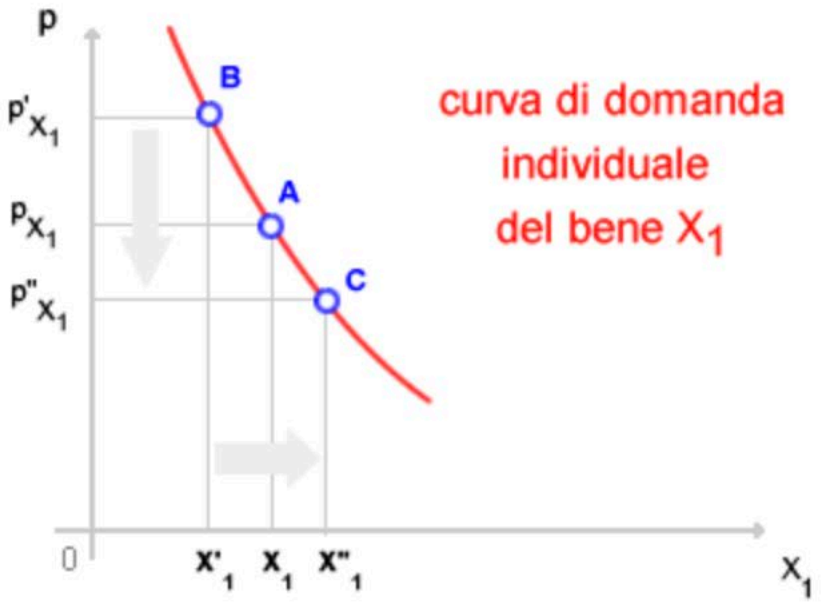
- Different consumers have different reservation prices for the same good.
- The reservation price guides purchasing decisions:
  - RP ≥ price set by producers → purchase
  - RP < price set by producers → no purchase
- **Knowing the reservation price makes it possible to construct the individual demand curve.**

The individual demand curve shows the quantity of the good demanded at each price by a consumer.

Individual demand curve for good x: **expresses, for each consumer, the reservation price for different quantities of x.**

The individual demand curve (1/2)

- The individual demand curve for good x expresses, for each consumer, the reservation price for different quantities of x.

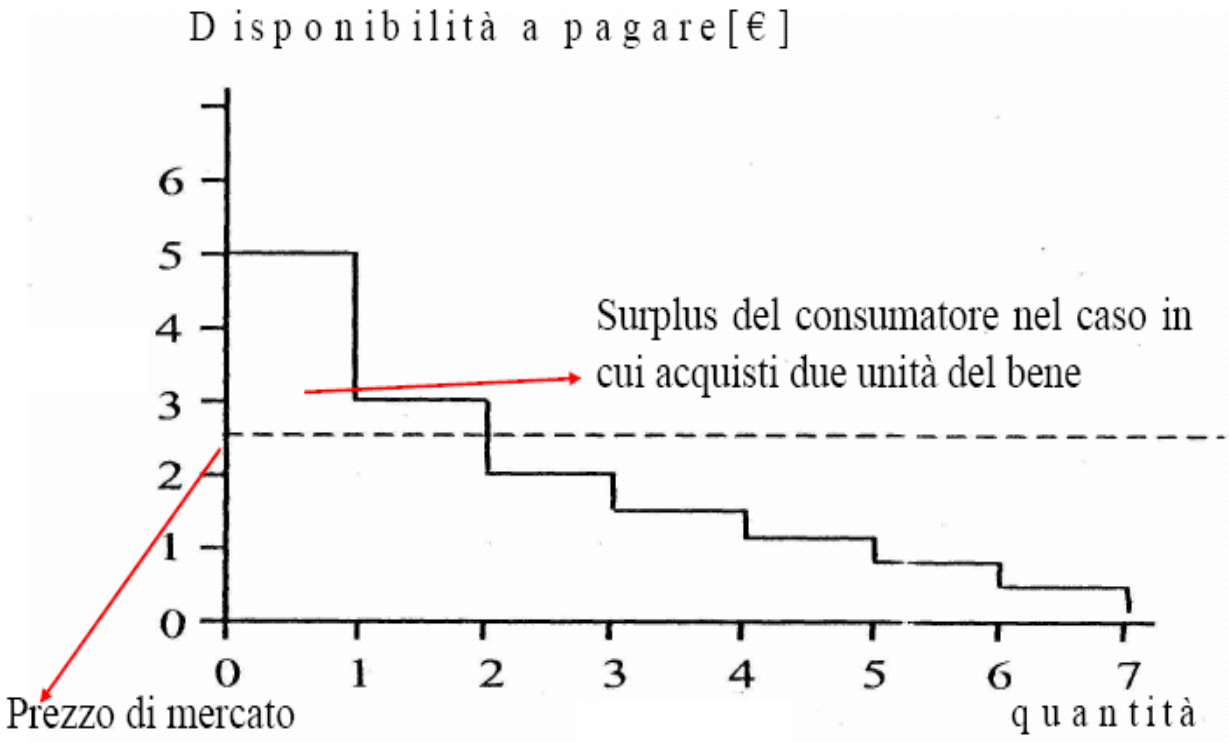


The individual demand curve indicates to the firm producing x:

- – given a price p, how many units (x) the consumer purchases
- – given the number of units the firm wants to sell to a consumer, what price p must be set for the consumer to buy them

The individual demand curve (2/2)

- The individual demand curve makes it possible to evaluate consumer surplus, defined as the difference between the price a consumer is willing to pay for a good and the market price of the good.
-



- **The individual demand curve is downward sloping.**

Why the curve is decreasing?

- The change in utility resulting from purchasing and consuming an additional unit of the good (marginal utility) **is decreasing.**
  - Therefore, the reservation price depends on the quantity of the good already owned or consumed

- The quantity demanded of a good varies inversely with its price, which holds for ordinary goods, that is, most goods.
  - Price increases → quantity demanded decreases
  - Price decreases → quantity demanded increases

Determinants of individual demand

Individual demand depends on:

Consumer characteristics
– tastes and needs
– income or wealth

Characteristics of the good
– price and availability of substitute goods
– price and availability of complementary goods

Determinants of individual demand: consumer characteristics

Consumers' tastes and needs (preferences).

Example: the quantity of Coca-Cola demanded by an individual depends on whether they like Coca-Cola and on their level of thirst.

Consumers' income or wealth.

For most goods (normal goods), the quantity demanded increases as income or wealth increases.

For some goods (inferior goods), the quantity demanded decreases as income or wealth increases (becasue they buy higher quality or luxury goods).

Determinants of individual demand: characteristics of the good

- Price and availability of substitute goods.
  - **Substitutes** for x are goods that perform similar functions to x.
  - Impact of substitutes goods on the demand for x
    - If the price of a substitute for x decreases ↓ → the quantity demanded of x decreases ↓ → as part of demand shifts to the substitute.
    - If the price of a substitute for x increases ↑ → the quantity demanded of x increases ↑ → the quantity demanded of x increases, not buying the substitute

- Price and availability of complementary goods.
  - **Complementary** goods are goods that tend to be consumed together.
  - Impact of complementary goods on the demand for x
    - If the price of a complement of x decreases ↓ → quantity demanded of x increases↑
    - If the price of a complement of x increases ↑ → quantity demanded of x decreases ↓

Individual and market demand

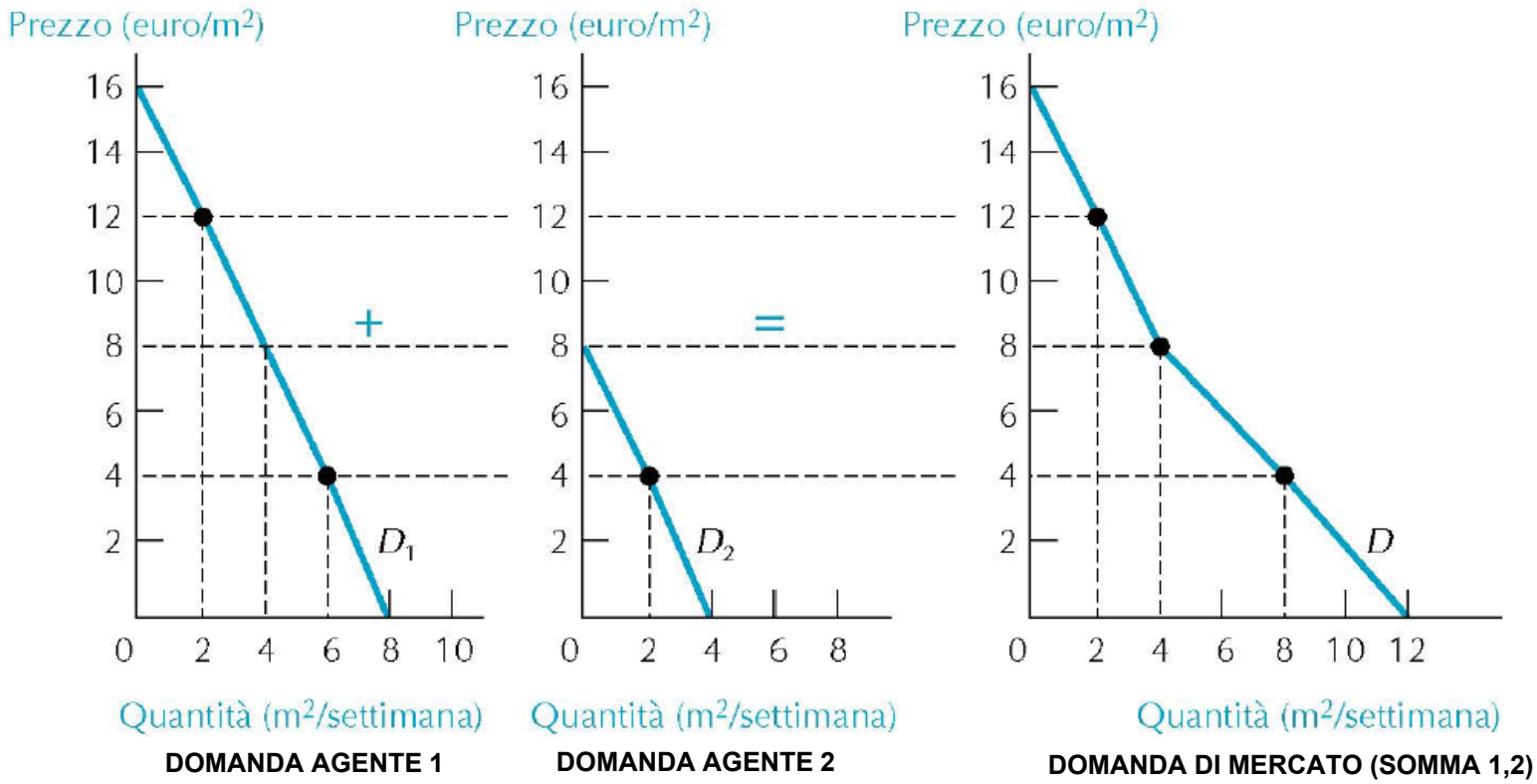
- The individual demand curve shows the quantity of the good demanded at each price by a single consumer.
- The market demand curve is obtained by summing individual demands at each price for all consumers in the market.



- **Market demand (aggregate demand) for good x is the sum of individual demands for x across all consumers.**

Demand curve

- The individual demand curve shows the quantity of the good demanded at each price by a single consumer.



- **It can take various functional forms, but it is always downward sloping.**
  - Simplifying assumption adopted here: linear demand curve.

Elasticity of the demand

- Elasticity of demand measures the change in quantity demanded resulting from a change in:
  - the price of the good.
  - the price of the good
  - income of individuals

Price elasticity of demand

- **Price elasticity of demand measures the percentage change in quantity demanded resulting from a one percent change in the price of the good.**

$$\varepsilon = \frac{dQ}{dp}\frac{p}{Q} = \frac{\frac{dQ}{Q}}{\frac{dp}{p}}$$

•Elasticity corresponds to the ratio between relative or percentage changes in quantity and price.

Price elasticity of demand

In a graphical representation of the demand curve, elasticity varies from point to point.
For this reason, point elasticity is usually calculated using infinitesimal calculus.

- **Point elasticity is equal to the first derivative of demand with respect to price.** Δq_x

$$\varepsilon_x = \frac{q_x}{\frac{\Delta p_x}{p_x}} \Rightarrow \varepsilon_x = \frac{\Delta q_x}{\Delta p_x} \cdot \frac{p_x}{q_x} \Rightarrow \varepsilon_x = \frac{\partial q_x}{\partial p_x} \cdot \frac{p_x}{q_x}$$

For ordinary goods, elasticity is negative.
In general, the absolute value is considered.

$$\varepsilon_x = \left| \frac{\partial q_x}{\partial p} \cdot \frac{p_x}{q_x} \right|$$

Price elasticity of demand

- **Example**

Example of demand elasticity with respect to cinema ticket prices.

$$\text{Demand Elasticity} = \frac{\%\ \text{Change in Quantity Demanded}}{\%\ \text{Change in Price}}$$

Initial ticket price: $10
New ticket price: $12
Initial quantity demanded: 200 tickets
New quantity demanded: 160 tickets

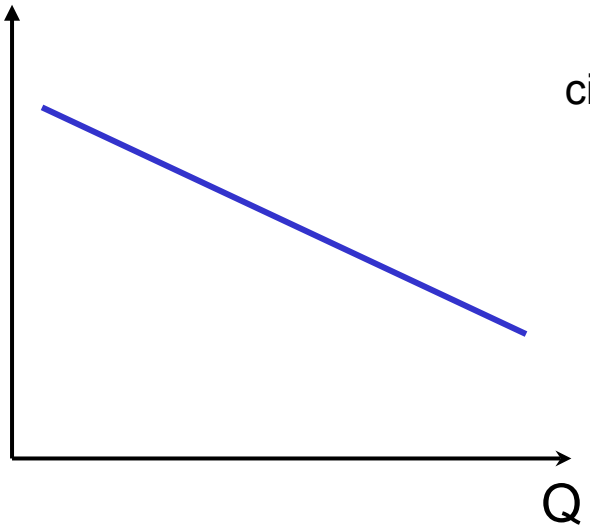
$$\text{Demand Elasticity} = \frac{-20\%}{20\%} = -1$$

Absolute value = 1
Curve: 45°

This indicates that demand for cinema tickets is elastic with respect to price.
In other words, a 20 percent increase in the ticket price leads to a 20 percent decrease in the quantity demanded.

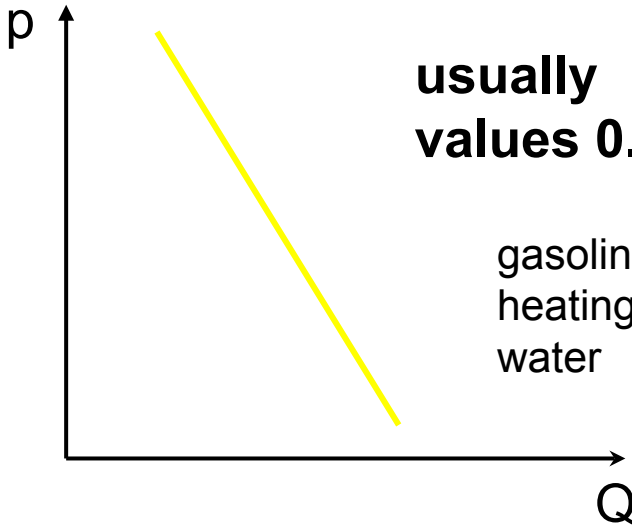
Demand curves and price elasticity

Elastic demand curve: $\varepsilon_x > 1$ p



cinema tickets

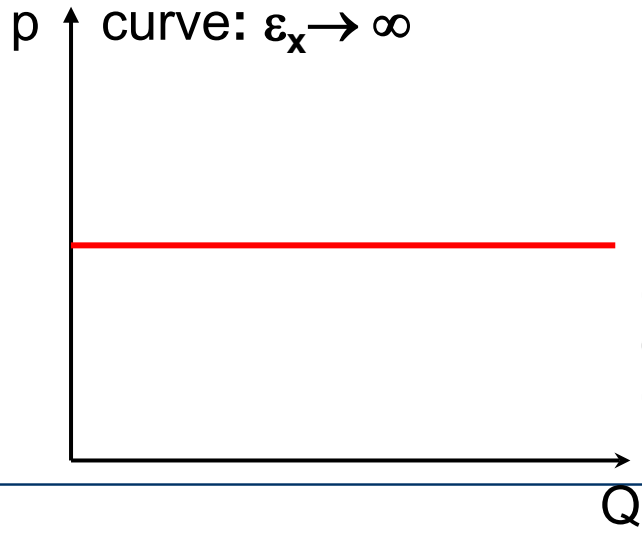
Inelastic demand curve: $\varepsilon_x < 1$



usually values 0.X

gasoline, heating, water

Perfectly elastic demand curve: $\varepsilon_x \rightarrow \infty$



highly substitues products in situation of competitions - brands of pasta, pepsi vs coke

Perfectly inelastic demand curve: $\varepsilon_x = 0$ p



unique products: life-saving medicals

Elasticity to the price

Demand for a good with few substitutes is relatively inelastic.
If price increases, it is difficult to find a substitute.
Examples include essential ordinary goods such as bread and gasoline

Demand for a good with many substitutes is highly elastic.
If price increases, consumers easily replace the good with another one.
Examples include specific types of wine, fruit, or cane sugar.

Income elasticity of demand

- Income elasticity measures the responsiveness of demand to changes in income.

$$\varepsilon_M = \frac{\partial q_x}{\partial M} \cdot \frac{M}{q_x}$$

It depends on the nature of the good.
For normal goods, income elasticity is positive.
For inferior goods, income elasticity is negative.

A special case: demand for goods in the presence of network externalities

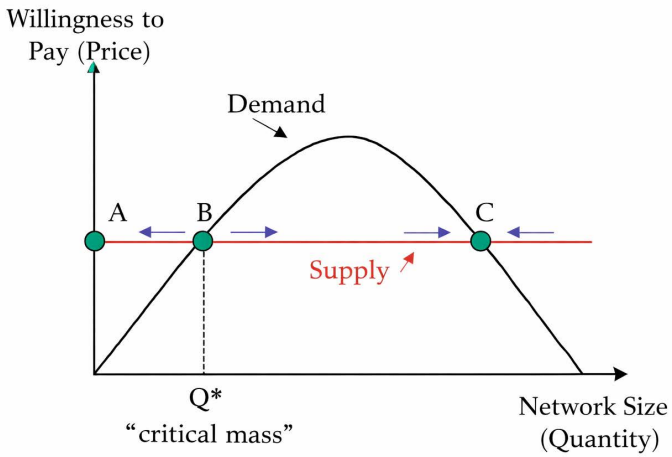
Network externalities

- **Network externalities**
  - Network externalities occur when the value of a good increases with the number of users or owners.
- Examples include telephone networks, email, hardware and software, and social events.
  - **Direct network externalities** arise when the value of the good automatically increases as the number of users increases, as with telephones or email.
  - **Indirect network externalities** arise when the benefit a consumer receives increases with the number of users due to a greater availability of complementary products.
-

Demand for a good in the presence of network externalities

- Demand is based on consumers' expectations regarding the size of the network.
  - The demand curve has an inverted U shape due to the interaction between network effects and price effects.
  - **When the network is small, few consumers have adopted the good and willingness to pay is low.**
  - **As the network grows, the value of the good and willingness to pay increase.**
  - When the network becomes very large, willingness to pay decreases because **remaining potential consumers have very low reservation prices.**

example: price of text messages





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The firm and the competition dynamics

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Who takes part of competition in the market?

Market: the place where exchanges between consumers and producers take place.

The set of consumers determines demand.

The set of producers determines supply.

Competitors: firms that produce goods perceived by consumers as substitutes.

Competing firms operate in the same market.

Substitutability: It depends on the characteristics of the good.

Would consumers want to substitute a Ferrari with a Smart?

It depends on geography.

Do consumers want and are they able to substitute products available in different national markets?

Do producers want and are they able to sell their products in markets located in different countries?

Quantity demanded is the quantity of a good that buyers wish to purchase. This quantity mainly depends on price.

- **The law of demand:** ceteris paribus, the quantity demanded of a good decreases as its price increases and increases as its price decreases.

Quantity supplied is the quantity of a good that sellers are willing to sell. This quantity mainly depends on price.

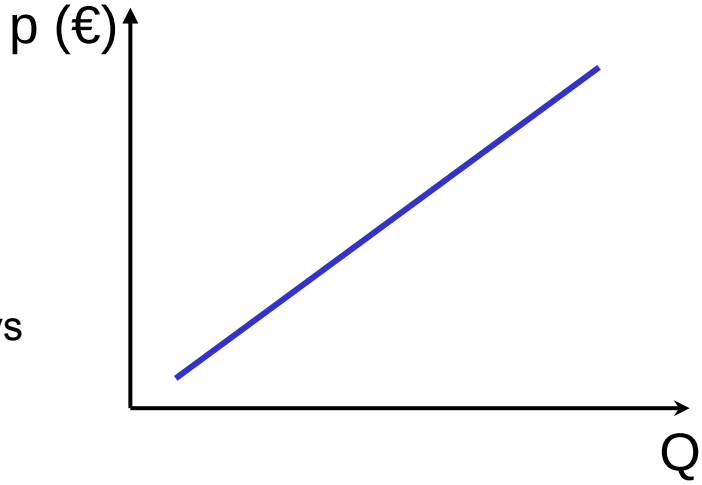
What happens to quantity supplied when price increases?

- **The law of supply:** ceteris paribus, the quantity supplied of a good increases as its price increases and decreases as its price decreases.

Supply

- A fundamental decision for firms is how much of a good to produce and sell.
  - Market supply for good x is the quantity of x that firms in a market are willing to produce and sell.
  - Market supply curve expresses, for each price level, the optimal quantity of production and sale of x.
- It is obtained as the sum of individual supply curves.

Assumption: marginal cost of production always increasing



It can take different functional forms, but it is always upward sloping.

- Simplifying assumption adopted here: linear supply curve.

Supply

As with demand, market supply coincides with the sum of the supplies of individual firms.

Market supply depends on:

- **production costs**, which depend on input costs and technological progress;
- **prices of related goods**;
- **regulatory policies**;
- **specific influences**.

- How can individual supply curves be constructed?
- **We start from costs.**

Construction of the individual supply curve

- Optimal production quantity of good x at price p is the quantity q that allows the firm producing x to maximise profit, defined as revenues minus costs.

$$\underset{q}{Max}\,\pi = p \cdot q - CT\,(q)$$

Construction of the individual supply curve

- Minimum production condition $\pi \geq 0$

$$p \cdot q - CT(q) \geq 0 \Rightarrow p \geq \frac{CT(q)}{q}$$

In general, not all costs depend on q.

CT(q)=CV(q)+CF, q≥ 0

- CV(q): variable costs, are the costs of productive inputs that vary with the level of output q.
- If production is zero, CV(0) = 0.
- CF: fixed costs, are costs that do not vary with q and are incurred even if production is zero.

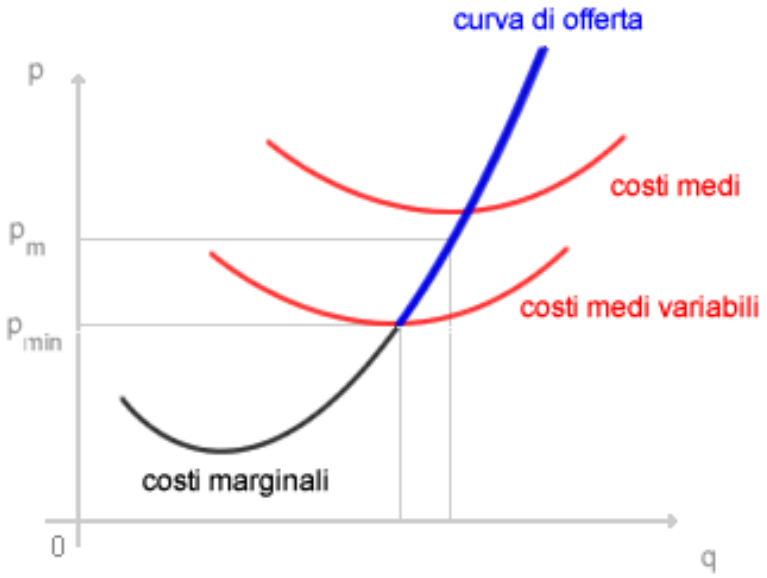
Without CF, the minimum production condition is:

$$p \cdot q - CV(q) \geq 0 \Rightarrow p \geq \frac{CV(q)}{q}$$

Average variable cost (AVC) = CV (q)/q

Construction of the supply curve in the short run

- In the short run, a firm's supply curve coincides with the portion of the marginal cost curve above average variable cost.

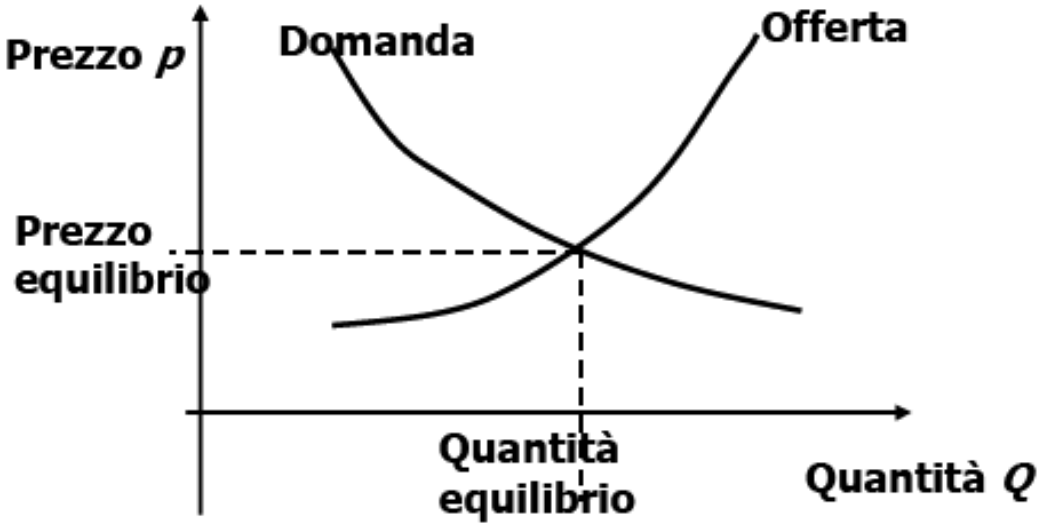


When the selling price is lower than average variable cost, no supply takes place because production would generate losses.

In the short run, fixed costs cannot be adjusted; therefore, firms focus on average variable cost.

How is the price determined?

- In a market with many producing firms, the equilibrium price depends on the interaction between demand and supply.



  - At the equilibrium price, there is an equilibrium level of production and exchange, known as the **equilibrium quantity.**

Why and how do equilibrium price and quantity emerge? (1/2)

- $P > P_{Equilibrium}$
  - $Q_{Demand} < Q_{Equilibrium} < Q_{Supply}$

    If price is higher than the equilibrium price: quantity demanded is lower than equilibrium quantity, which is lower than quantity supplied.

    There is at least one firm that does not sell but could offer the good at a lower price.

→ This firm negotiates a price lower than P with some unsatisfied consumers. < P

→ Other firms will adjust to the lower price.

→ The process is repeated until equilibrium is reached.

Excess supply lead to price reduction until equilibrium

Why and how do equilibrium price and quantity emerge? (2/2)

- $P < P_{Equilibrium}$
  - $Q_{Demand} > Q_{Equilibrium} > Q_{Supply}$
  - If price is lower than the equilibrium price: quantity demanded is higher than equilibrium quantity, which is higher than quantity supplied.

Some consumers are not served but are willing to pay a higher price.

→ Consumers negotiate a higher price with a firm in a mutually beneficial way > P

→ Competing firms, if any, will also adjust their prices upward.

→ The process is repeated until equilibrium is reached.

Excess demand will lead to higher price adjustment

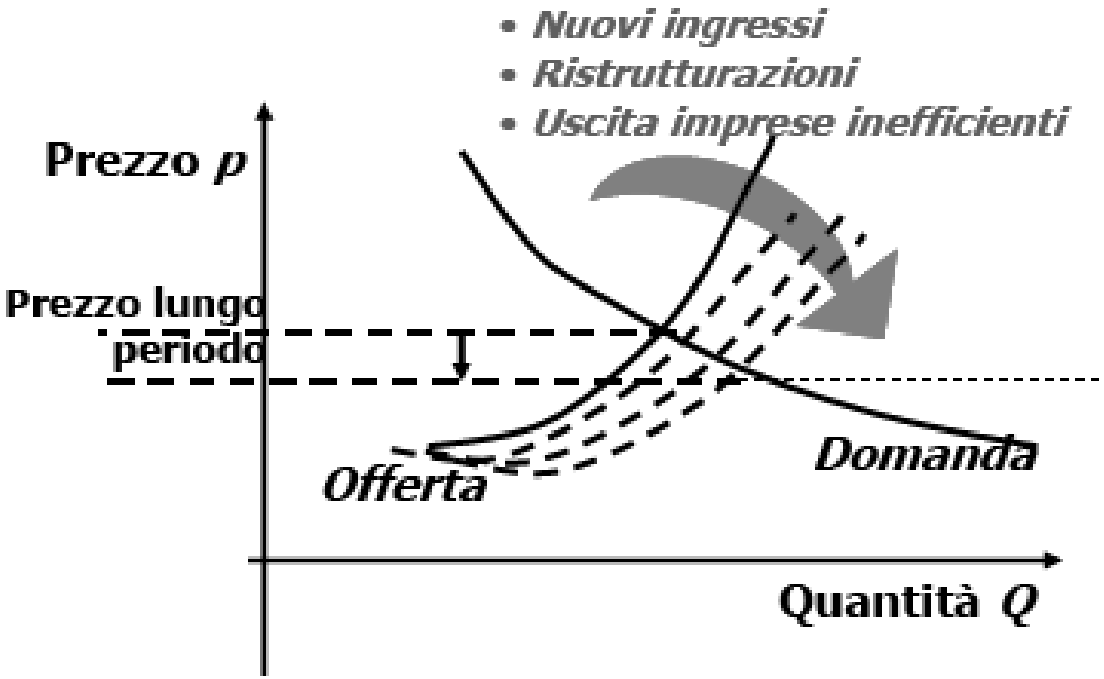
What happens over time?

- If $P_{Equilibrium}$> Average cost, firms earn profits
  → New firms enter the market attracted by profits.
  → demand ↑ and price ↓
  → for some firms p < average cost
  → firms with average costs > p exit the market
  → supply ↓ e price ↑
  …. and so it goes on

- **Long term equilibrium** is reached when entry and exit stop because no further profits are possible (perfect competition lead to profits=0)
  Average cost=Average $cost_{min}$
  price=Average $cost_{min}$
  Average $cost_{min}$: average cost of most efficient firms

Long term equilibrium

The long-run supply curve shifts downward until it becomes horizontal at the level of minimum average cost



Underlying assumptions

- All these results hold only if certain conditions are satisfied.

- Technological symmetry: all firms have access to the same technology under the same conditions, and the best technologies can be imitated at no cost.
- Free entry: new firms can enter the market at the same cost as existing firms, so imitation is also possible for firms not yet operating in the market.
- Free exit: firms can exit the market without incurring additional costs, so they do not remain trapped in loss-making industries.

- **These conditions characterise the market structure known as perfect competition.**

Perfect competition

- A perfectly competitive market consists of firms producing identical goods and selling them at the same price.
- The output of each firm is so small relative to total market demand that it cannot influence the market price.
- The quantities purchased by each buyer are so small that they do not significantly affect the market price.
- The quantities of inputs purchased by each producer are so small that they do not affect input prices.

Perfect competition

- Firms produce homogeneous goods that consumers perceive as identical.
- Consumers have perfect information about the prices of goods available on the market.

- All firms, both incumbents and potential entrants, have equal access to resources such as technology and inputs.

Implications

- **Price-taker behaviour:** buyers and sellers take the product price as given when deciding how much to buy or produce.
- **Law of one price:** there is a single price at which transactions occur.
- **Freedom of entry:** if new firms find it profitable to enter the industry, they are free to do so.